

Kapa HyperPlus DNA Prep With PCR

Prepared by:	Robert Cormier	2024-06-24
Revised by:	Alex-J Roussel	2025-09-02
Approved by:		

Version	Enforcement Date
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REVISIONS FROM THE ORIGINAL VERSION

Version	Summary of modifications	Date and initials
	Fragmentation time for high quality DNA, such as buffy coat samples, should be 18 minutes	02/04/2025 - AJR
	Minor changes to make the protocol more concise	02/09/2025 - AJR

Date:	
Project:	
Person in charge	
Lib prepared by:	

Goal: This protocol uses the KAPA Hyperplus kit to fragment and convert genomic DNA extracted from tissue or FFPE into low-biased, representative libraries for sequencing using Illumina Novaseq Platform.

Materials required:

Item	Source
KAPA Fragmentation Enzyme	
KAPA Fragmentation Buffer (10X)	
Conditioning Solution	
End Repair & A-Tailing Enzyme	
End Repair & A-Tailing Buffer	
Ligation Buffer	
DNA Ligase	
KAPA Hifi HotStart Readymix (2X)	
Library Amplification Primer Mix (10X)	
10 mM Tris-HCl (pH 8.5)	
96 well PCR Plate Non-skirted	(Phenix Research, Cat # MPS-499) or similar)
Thin-wall nuclease-free PCR tubes	
Thermocycler	
Heat block	
Magnetic Stand -	96 (Ambion, Cat # AM10027) or similar

KAPA Hyperplus DNA Prep With PCR

NOTE: Remove enzymes from -20 immediately before adding to master mix, flick to mix + 600 x g quickspin, and immediately put back.

NOTE: Vortex and centrifuge reagents (except enzymes) in a microcentrifuge before use to ensure material has not lodged in the can or side of the tube.

NOTE: When mixing reactions by pipetting using the multichannel, increase the total volume on the pipette to ensure adequate mixing. Undermixing will result in failure of the library prep.

NOTE: For all Clean-up steps, pre-aliquote reagents in PCR strips to allow use of multichannel.

NOTE: Do not freeze Ampure XP Cleanup Beads. Beads should be stored at 4°C and brought to room temperature before use.

NOTE: Library construction workflows must be tailored and optimized to accommodate specific experimental designs, sample characteristics, sequencing applications, and equipment. For additional information, consult the KAPA NGS Library Preparation Technical guide, or visit sequencing.roche.com/support.

NOTE: DNA resuspended in buffer containing EDTA must first be treated with 5 µL conditioning solution during dilution. Failure to treat with conditioning solution will result in inhibition of fragmentation enzyme, which can affect fragmentation efficiency.

DNA Input Recommendation

Complex gDNA (high quality) – 50ng -1µg

Target capture (WES, custom panels) Complex gDNA (high quality) – 10ng – 1 µg

WGS, target capture – FFPE DNA - ≥ 50ng (quality dependant)

WGS, microbial DNA - 1ng – 1 µg

Targeted sequencing, long amplicons - ≥1 ng

RNA-seq, Full-length/unfragmented cDNA - ≥1 ng

Step A: Enzymatic Fragmentation of Genomic DNA

Required Reagents:

- KAPA Fragmentation Buffer (10X) – 5µL/Reaction
- KAPA Fragmentation Enzyme – 10 µL/Reaction

NOTE: KAPA Fragmentation buffer and enzyme can be combined on ice prior to addition.

- ☐ **Thaw whole Kit in 4°C Freezer. (Store reagents at -20°C after use)**
- ☐ IF THE SAMPLES FAILED NANODROP QC, perform a pre library clean-up with a 3:1 Ampure Beads to Starting volume ratio.
- ☐ Dilute the DNA samples in 10 mM Tris HCl (pH 8.5) to a final volume of 35 µl in a PCR plate and keep on ice.
- ☐ Preheat the thermocycler to **Enzymatic Fragmentation** (with lid turned off)
- ☐ Prepare a master mix using the following reagents on ice according to the following table and mix by pipetting:

Reagent (Lib Prep Kit, -20°C)	Volume/sample
DNA input	35 µL
KAPA Fragmentation Buffer (10X)	5 µL
KAPA Fragmentation Enzyme	10 µL
Total	15 µL /well of MasterMix; 50 µL total

- ☐ Add 15 µl of master mix to each well containing the DNA input and pipette to mix.
- ☐ Place the plate into the preheated thermocycler, perform the following cycling conditions:
Enzymatic Fragmentation
- ☐ Take out the buffer mix for Step B: End Repair & A-Tailing and thaw at room temperature.

NOTE: Fragmentation time : Buffy Coat is 20 minutes, FrzT is 18 minutes and FFPE is 15 minutes.

Temperature	Duration	
4°C	∞	☒
37°C	15 minutes	☒
4°C	∞	☒

- ☐ Spin plate and transfer reactions to ice. Pre-heat Thermocycler to **End Repair & A-Tailing** (with lid at 85°C).
- ☐ Proceed immediately to Step B: **End Repair & A-Tailing**

Step B: End Repair & A-Tailing

Required Reagents:

- KAPA End Repair & A-Tailing Buffer – 7 µL/Reaction
- KAPA End Repair & A-Tailing Enzyme Plus Mix – 3 µL/Reaction

- ☐ Prepare the following reagents on ice (except the fragmented DNA) and mix by pipetting until visibly homogenous.

Reagent (Lib prep Kit, -20°C)	Volume/ 1 sample	*X	
Fragmented Double-Stranded DNA	50µl	--	☒
End Repair & A-Tailing Buffer	7 µl		☐
End Repair & A-Tailing Enzyme Plus Mix	3 µl		☐
Total	10 uL/well mastermix; 60 µl Total		

- ☐ Add 10 µl of the End Repair & A-Tailing Mastermix into each sample well and pipette to mix.
- ☐ Spin plate and immediately place the plate into the preheated thermocycler, perform the cycling conditions: **End Repair & A-Tailing**

Temperature	Duration	
65°C	∞	☒
65°C	30 minutes	☒
20°C	∞	☒

Note: Both the fragmentation enzyme and the end repair enzymes are inactivated at 65°C. When reactions are set up according to recommendations, additional fragmentation should be negligible. The brief period of end repair is sufficient for enzymatically fragmented DNA.

- ☐ Take out reagents for **Step C: Adapter Ligation** near the end of the incubation.
- ☐ Following incubation, immediately proceed to Step C: **Adapter Ligation**

Step C: Adapter Ligation

Required Reagents:

- Adapter Stock Solution – 5 µl/Reaction
- Ligation Buffer – 30 µl/Reaction
- DNA Ligase – 10 µl/Reaction
- Nuclease Free Water - 5 µl/Reaction

NOTE: Ensure to mix the ligation reaction until visibly homogenous by vortexing. Reagents are viscous and there is little chance of overmixing. Failure to mix adequately will result in failure of library prep.

- ☐ Take out Ampure XP beads and bring to room temp (30 mins)
- ☐ Preheat thermocycler (Adapter Ligation)
- ☐ Add 5 µl of adapter stock to the End Repair & A-Tailing Reaction Product (65 µl Total).
- ☐ Prepare the following reagents on ice (except the fragmented DNA + Adapter) and mix by pipetting until visibly homogenous.

Reagent (Lib Prep Kit, -20°C)	Volume/ 1 sample	*X	
End Repair & A-Tailing Reaction Product + Adapter Stock	65 µl	--	☒
PCR-Grade water*	5 µl		☒
Ligation Buffer*	30 µl		☒
DNA Ligase*	10 µl		☒
Total	45uL/well mastermix; 110uL Total		

- ☐ Add 45 µl Adapter Ligation MasterMix to each sample, Mix thoroughly and spin plate.
- ☐ Incubate the reactions following the conditions below: (Adapter Ligation)

Temperature	Duration	
20°C	∞	☒
20°C	15 minutes	☒

- ☐ Prepare 80% EtOH and aliquot 10mM Tris-HCl, pH 8.5 during this incubation.
- ☐ Proceed immediately to Step D: Post Ligation Cleanup

Step D: Post-Ligation Cleanup

- ☐ Thoroughly vortex Ampure XP beads to ensure complete resuspension.
- ☐ Add 88 µl Ampure XP beads to the samples, pipette to mix.
- ☐ **Incubate plate at room temperature for 5-15 minutes.**
- ☐ Place the plate on a magnet for 10 minutes (or until liquid is clear)
- ☐ **Thaw reagents for Step E: Library Amplification on ice.**
- ☐ **Preheat Thermocycler to “Library Amplification”**
- ☐ REMOVE AND DISCARD THE SUPERNATANT.
- ☐ Add 200 µL of freshly prepared 80% EtOH, incubate for 30 seconds, and remove ALL of the supernatant.
- ☐ Repeat the previous step. (Total : 2 wash)
- ☐ Air-dry sample for 3 minutes. After one minute, remove ALL residual liquid that may have collected at the bottom of the well, using a 10 µl multi-channel.
- ☐ **Remove plate from magnetic stand** and resuspend bead pellet in 25 µL of 10 mM Tris-HCl, pH 8.5 by pipetting volume up and down. Ensure that beads are completely resuspended.
- ☐ Incubate for 2 minutes.
- ☐ Incubate the sample on the magnet for 3 minutes or until solution is clear.
- ☐ Transfer 20 µL (2 x 10 µl) of supernatant to a new well of a PCR plate.
- ☐ Immediately proceed to **Step E: Library Amplification**

POST-PCR ROOM

Step E: Library Amplification

Required Reagents

- KAPA HiFi HotStart ReadyMix (2X) - 25 µl/Reaction
- Library Amplification Primer Mix (10X) – 5 µl/Reaction

Depending on initial DNA input, select the proper amount of PCR cycles:

Input into library construction	PCR cycles required to generate	
	100 ng library	1 µg library
1 µg	0	0-1
500 ng	0	2-3
250 ng	0-1	3-5*
100 ng	0-2	5-6
50 ng	3-5	7-8
25 ng	5-6	8-10
10 ng	7-9	11-13
5 ng	9-11	13-14
2.5 ng	11-13	14-16
1 ng	13-15	17-19

☐ **On ice, add the following reagents to the samples in order:**

Reagent	Volume/ sample	
Adapter-Ligated Library	20 µl	☒
1 st : Library Amplification Primer Mix (10X)	5 µl	☒
2 nd : KAPA HiFi HotStart ReadyMix (2X)	25 µl	☒
Total	50 µl	

☐ Mix well by pipetting up and down (total 50uL). Spin Plate

☐ Place the tubes into a thermocycler and, with a heated lid, perform the following cycling conditions depending on the initial input, needed outcome, and origin of the sample:

Temperature	Duration	Cycles
98°C	45 seconds	☒
98°C	15 seconds	☐ 4* *Number of cycles varies based on the original DNA input amount
60°C	30 seconds	
72°C	30 seconds	
72°C	1 minutes	☒
4°C	∞	☒

- ☐ Take out Ampure XP beads and bring to room temp (30 min)
- ☐ Prepare 80% EtOH and 10 mM Tris-HCl, pH 8.5
- ☐ Proceed directly to **Step F: Post-Amplification Cleanup**

Step F: Post-Amplification Cleanup

- ☐ Take out the D1000 screen tape and reagents, and Qubit dsDNA BR kits and allow to sit at room temperature for 30 minutes before use.
- ☐ Prepare 80% EtOH (400 µl/sample), and 10 mM Tris-HCl, pH 8.5
- ☐ Add 50 µL Ampure XP Beads to the samples and mix well by pipetting.
- ☐ Incubate for 5-15 minutes at room temperature.
- ☐ Incubate the sample on the magnet for 10 minutes or until solution is clear.
- ☐ REMOVE AND DISCARD THE SUPERNATANT
- ☐ Add 200 µL of freshly prepared 80% EtOH, incubate for 30 seconds, and remove ALL of the supernatant.
- ☐ Repeat the previous step. (Total : 2 wash)
- ☐ Air-dry sample for 3 minutes. After one minute, remove ALL residual liquid that may have collected at the bottom of the well, using a 10 µl multi-channel.
- ☐ Remove plate from magnetic stand and resuspend bead pellet in 22 µL of 10 mM Tris-HCl, pH 8.5 by pipetting volume up and down. Ensure that beads are completely resuspended.
- ☐ Incubate for 2 minutes.
- ☐ Incubate the sample on the magnet for 3 minutes or until solution is clear.
- ☐ Transfer 20 µL (2 x 10 µl) of supernatant to a new PCR plate.
- ☐ Seal the plate, shake at 1800rpm for 1 min then quick spin ☐ QC the Final Library:
- ☐ Check the size distribution of the final library by TapeStation D1000 Assay (Agilent)
- ☐ Quantify library using Qubit dsDNA BR kit.

- ☐ Add Lib Prep Plate to Plate Inventory & Update which indexes were used in the Index Inventory